A Human-in-the-Loop LLM Workflow for Research Roadmap Curation

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Abstract

Early-stage research often begins with vague ideas that are not yet executable. Converting a broad interest into a concrete project requires research questions, literature directions, datasets, baselines, evaluation metrics, ablations, milestones, risks, and a realistic contribution claim. Large language models (LLMs) can assist with this planning stage, but unstructured prompting tends to produce roadmaps that are fluent yet generic, hard to verify, or poorly scoped. We present a human-in-the-loop LLM workflow for research roadmap curation. Given a raw research idea, the workflow generates a structured roadmap artifact that is reviewed, revised, and approved by a human reviewer, who may be the researcher themselves, a mentor, an instructor, a collaborator, or a domain expert. The workflow is model-agnostic; our prototype conditions an LLM backend on reference templates, optionally retrieves relevant literature, emits LaTeX, compiles a PDF, and exposes the result for review through an asynchronous job queue. We evaluate the workflow with a blind human preference study in which 40 evaluators rated four anonymized roadmap variants for the same idea on completeness, feasibility, specificity, experiment design, and research potential. Of the 40 evaluators, 34 (85%) selected the roadmap produced by our workflow as the one they would adopt overall, versus 6 combined for the three unstructured ChatGPT baselines. The result is consistent with the hypothesis that schema-constrained generation under human review yields more usable first-pass research plans than one-shot prompting, while keeping a human accountable for feasibility and novelty.

1. Introduction

Early-stage research projects often fail before any experiment is run. The initial idea is too broad, the dataset is unclear, the baselines are missing, the evaluation metrics do not match the question, or the expected contribution is overclaimed. This pattern is especially common in research bootcamps, undergraduate research programs, industry training cohorts, and early-stage graduate projects, where participants begin with enthusiasm but limited experience converting a broad interest into an executable plan.

A vague idea is not yet a research project. Statements such as “I want to work on multimodal RAG”, “I want to apply reinforcement learning to retrieval”, or “I want to use scientific machine learning for physical systems” are useful starting points, but they do not specify what should be built, what should be compared, what evidence should be collected, or what claim the final paper or report could defend. To become actionable, such ideas must be transformed into structured roadmaps that include research questions, literature directions, datasets, baselines, metrics, ablations, weekly milestones, risks, deliverables, and a minimum viable claim. This transformation is typically performed by a human mentor, instructor, supervisor, or domain expert, and it does not scale: in a cohort with many students, mentors must repeatedly convert loosely stated interests into detailed project plans, and the bottleneck is not the document itself but the judgment required to keep it specific, feasible, experimentally sound, and appropriately scoped.

Large language models (LLMs) are a natural fit for this stage. However, one-shot prompting is often insufficient. A generic prompt such as “generate a 12-week research roadmap for this idea” produces fluent output that may omit baselines, suggest unrealistic datasets, fail to define measurable success criteria, or overstate the novelty of the project. The surface fluency of LLM-generated prose can also make weak plans appear more

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credible than they are. Our central question is therefore: can a structured, human-reviewed LLM workflow help convert vague research ideas into executable roadmaps more reliably than unstructured prompting?

We present a human-in-the-loop LLM workflow for research roadmap curation. Given a raw research idea, the workflow generates a structured roadmap artifact, which is then routed to a human reviewer who can inspect, revise, or approve it. The reviewer may be the original researcher, a mentor, an instructor, a supervisor, a collaborator, or a domain expert. The workflow is model-agnostic: it can be instantiated using API-based models, coding agents, or local models, as long as the backend supports structured generation and artifact creation. Our prototype processes requests asynchronously, conditions on reference templates, optionally retrieves relevant literature, emits LaTeX, compiles a PDF, and exposes the result for review.

We evaluate the workflow with a blind human preference study. For one research idea, four anonymized roadmap variants were generated: variants A, B, and D from three different unmodified ChatGPT versions prompted in an unstructured way, and variant C from our workflow. Forty evaluators rated all four roadmaps along five dimensions (completeness, feasibility, specificity and actionability, experiment design quality, and research potential) and recorded a single overall preference. 34 of 40 evaluators (85%) selected variant C overall, with the remaining 6 (15%) distributed across the three unstructured baselines.

We make three contributions:

- We formulate research-roadmap generation as a human-in-the-loop workflow rather than a one-shot generation task, and articulate why the human stays in the loop after generation rather than only at prompt time.
- We introduce a roadmap schema that decomposes early-stage research planning into twelve concrete components, including research questions, datasets, baselines, metrics, ablations, risks, milestones, and a minimum viable claim, so that quality stops depending on surface fluency.
- We describe and run a blind human-preference evaluation protocol that compares four anonymized roadmap variants for a shared research idea across five quality dimensions, and report a strong overall preference for the schema-constrained, human-reviewed variant.

The rest of the paper is organized as follows. Section 2 positions the workflow against related work on LLM-assisted research and structured prompting. Section 3 describes the workflow, the roadmap schema, and the asynchronous job lifecycle. Section 4 defines the evaluation protocol and study setup. Section 5 reports the preference results, and Sections 6 and 7 discuss implications and threats to validity.

2. Related Work

LLMs as research assistants and autonomous scientists. A growing body of work investigates large language models in roles ranging from open-ended idea generation and hypothesis proposal to end-to-end autonomous scientists that orchestrate experiments and write papers (Lu et al., 2024; Baek et al., 2025; Yang et al., 2024). These systems are ambitious about scope, often aiming to produce or evaluate full research outputs. Their evaluations focus on the quality of the produced artifact (a paper, a hypothesis, or an experiment) rather than on the planning artifact that precedes execution. Our framing is narrower and more operational: the LLM is a planning assistant that drafts an executable roadmap, and the human keeps responsibility for feasibility, novelty calibration, and scientific correctness.

Structured and schema-constrained generation. Prompting strategies that impose structure on LLM output (chain-of-thought (Bosma et al., 2022), self-consistency (Wang et al., 2022), and tree-of-thoughts (Yao et al., 2023)) consistently improve reliability on reasoning-heavy tasks. Schema- or grammar-constrained decoding (Willard & Louf, 2023; Geng et al., 2023) goes further by forcing outputs into a parseable format. Our roadmap schema is a coarse-grained instance of the same idea applied to research planning: rather than constrain syntax, we constrain the set of components a roadmap must populate (questions, datasets, baselines, metrics, ablations, milestones, risks, deliverables, minimum viable claim) so that quality stops depending on surface fluency.

Human-in-the-loop and mixed-initiative AI. Mixed-initiative systems with explicit human-AI interaction have a long history in HCI (Horvitz, 1999; Amershi et al., 2019). Recent work specifically on human-in-the-loop LLM workflows studies prompt iteration (Zamfirescu-Pereira et al., 2023), co-writing and editing assistants (Lee et al., 2022; Yuan et al., 2022), and AI-assisted code review and refactoring (Mozannar et al., 2024). The shared lesson is that the human contribution is not just the initial prompt but the editorial judgment applied after generation. Our workflow exposes this point of judgment explicitly through a reviewable artifact and an approve/revise state transition, rather than relying on free-form chat turns.

Human preference evaluation of LLM output. Pairwise and small-batch human preference judgments are now a standard evaluation primitive for LLM outputs, both for alignment training (Ouyang et al., 2022; Bai et al., 2022) and for benchmarking deployed models (Régin et al., 2024; Zheng et al., 2023). We adopt the same primitive (blind human selection among anonymized variants) but apply it to the relatively unstudied artifact of a research roadmap, and we ask evaluators to rate five quality dimensions chosen specifically for executability rather than for fluency or helpfulness.

Existing structured-planning artifacts. The twelve-component schema is not derived from a vacuum; it is closest in spirit to mentor checklists used in research bootcamps, to proposal-style templates such as the NeurIPS Datasets and Benchmarks checklist and ML reproducibility checklists (Pineau et al., 2021), and to the proposal scaffolds embedded in modern note-taking and project management tools (Notion AI templates, Airtable research planners, and similar). These artifacts predate the workflow described here and already encode many of the same components (research questions, datasets, baselines, milestones, deliverables). Our contribution relative to that body of practice is twofold: we bind the schema to a generation backend so the components are populated by an LLM rather than left as empty fields, and we put the populated artifact through an explicit reviewable revision loop. The schema itself overlaps substantially with prior planning rubrics; the workflow surrounding it is what is new.

Distinguishing this work. The closest prior art is the union of structured prompting, human-in-the-loop writing assistants, and existing research-planning checklists and templates. Structured prompting work shows that constraining the output space helps; human-in-the-loop writing work shows that retaining a human review step matters; autonomous science work shows that LLMs can produce research-shaped artifacts. None of these threads directly studies the early-stage planning artifact, namely an executable research roadmap, as a deliverable in its own right. Our contribution is to specify the schema, build a review-and-revision workflow around it, and evaluate the resulting roadmaps with a blind human preference study against unstructured prompting baselines.

3. Method

We define a research roadmap as an execution-oriented planning artifact that transforms a broad research idea into a concrete project plan. A good roadmap should help a researcher understand what to read, what to implement, what to compare, how to evaluate progress, and what claim the final project may support. Unlike a literature review or a project proposal, a roadmap is explicitly operational: it should describe both the intellectual structure of the project and its execution structure, including what needs to happen in week one, week two, and so on.

The central design choice is to separate generation from approval. The LLM drafts the roadmap; a human reviewer decides whether the roadmap is usable, and revises it through a feedback loop until it is. This loop is what distinguishes the workflow from one-shot prompting.

3.1. Workflow Overview

The workflow has five stages, illustrated in Figure 1: idea submission, structured generation, artifact creation, human review, and revision-or-delivery. A user submits a raw research idea together with optional constraints (target duration, researcher background, preferred domain, tool or dataset constraints, and a target output type such as paper, report, codebase, or presentation). An LLM backend generates a roadmap conditioned on a fixed schema (Section 3.2) and reference templates. The roadmap is converted into a reviewable artifact, in our prototype a LaTeX-source-plus-PDF document. A human reviewer inspects the artifact and either approves it or returns it with revision requests. On revision, the workflow regenerates or edits the roadmap with reviewer feedback in context; on approval, the roadmap is delivered or used directly.

The reviewer is not only the person who writes the initial prompt; the human remains involved after generation, when quality judgments are required. This makes the workflow human-in-the-loop rather than merely human-initiated. The reviewer may be the original researcher, a faculty advisor, a bootcamp mentor, a teaching assistant, a lab lead, or a domain expert. The core requirement is not a specific role but a decision point where a human accepts responsibility for the final plan.

3.2. Roadmap Schema

The LLM is not asked to freely brainstorm. Each generated roadmap is constrained to populate a fixed set of components, grouped into three families (Figure 2).

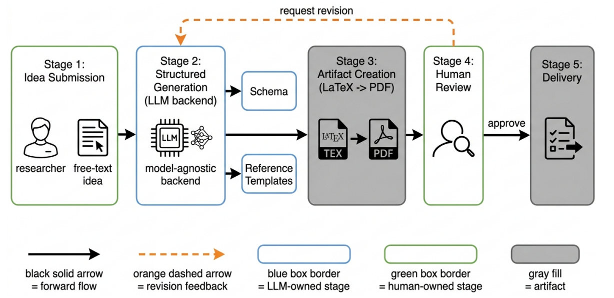

Figure 1. End-to-end human-in-the-loop roadmap-curation workflow. A raw research idea flows through schema-constrained LLM generation, artifact creation, and human review. The reviewer can approve the roadmap for delivery or route it back through the revision loop (orange dashed arrow). The human reviewer remains responsible for feasibility, novelty calibration, and scientific correctness.

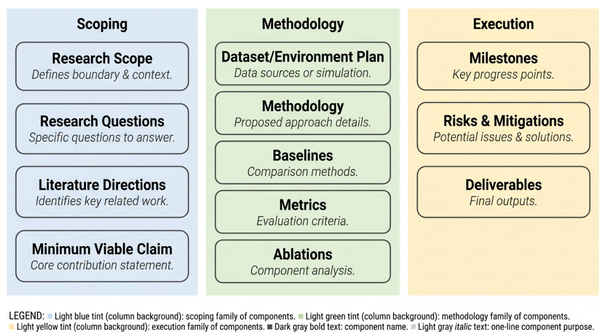

Figure 2. Twelve required components of the roadmap schema, organized into a scoping family, a methodology family, and an execution family. Schema-constrained generation forces the LLM to populate every component rather than producing free-form prose.

The scoping family defines what the project is and is not: research scope, research questions, literature directions, and a minimum viable claim. The methodology family defines what evidence the project will use and how the technical approach will be evaluated: dataset or environment plan, methodology, baselines, metrics, and ablations. The execution family converts the project into operational steps: weekly milestones, risks and mitigations, and concrete deliverables.

Reference templates are concrete instances of past roadmaps that follow the schema and ship with the workflow; the LLM backend is shown one or more matching templates as few-shot exemplars before it drafts a new roadmap, which steers the surface form toward the schema without dictating the substantive content. The schema itself is intentionally coarse-grained. It does not prescribe how each component is written; it only requires that each component be present and substantive. The twelve components and three-family

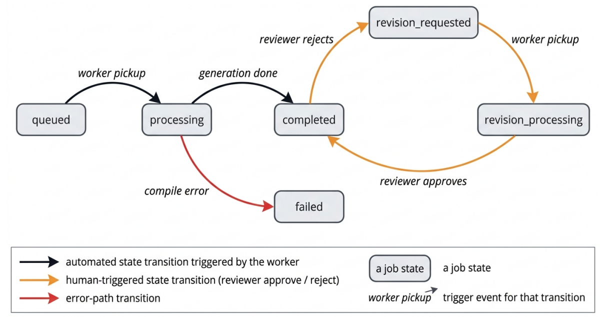

Figure 3. Job state machine for a roadmap request. The linear path queued → processing → completed is automated. The cycle completed → revision_requested → revision_processing → completed is the explicit human-in-the-loop hook that distinguishes the workflow from one-shot chat.

grouping are a design choice distilled from the mentor checklists used in the bootcamp from which the evaluator cohort is drawn, rather than a derivation from prior planning literature. The intent is to make roadmap quality less dependent on surface fluency. A roadmap is not strong merely because it reads well; it must specify how the project can actually be executed and evaluated. Concretely, the schema makes weak baselines, missing metrics, and unstated risks visible to a reviewer at a glance, because the slot for that component is empty or visibly thin.

3.3. Model-Agnostic Asynchronous Backend

The workflow is independent of any single LLM provider. The backend can be any model or agent that supports structured generation and optional tool use, including API-based models, coding agents, or local models. Our prototype uses an asynchronous worker architecture: a roadmap request is stored as a job, picked up by a worker that calls the LLM backend with the schema and templates, optionally retrieves relevant literature, emits LaTeX source, and compiles a PDF that is exposed to the reviewer.

The job state machine (Figure 3) supports six states: queued, processing, completed, failed, revision_requested, and revision_processing. The presence of a dedicated revision_requested state, separate from processing, makes the human review point a first-class concept rather than a chat turn. Long-running generation, compile errors, and asynchronous revisions all become observable transitions rather than hidden behavior. The job queue is not a scientific contribution by itself; it is the implementation mechanism that lets the workflow behave like a real review-and-revision system instead of a synchronous chat interaction.

3.4. Human Review Criteria

Human review is necessary because roadmap quality depends on judgments that LLMs cannot reliably make alone. A roadmap may appear polished while still being unrealistic, too broad, weakly evaluated, or scientifically shallow. The reviewer is asked to check whether: the research question is precise, the proposed dataset is accessible, the baselines are appropriate, the metrics are measurable, the milestones are realistic, the risks are honestly stated, and the minimum viable claim is not overconfident. If any of these checks fails, the reviewer moves the job into revision_requested with feedback; the feedback is provided as context to the next generation or editing cycle, and the loop repeats until the reviewer approves the roadmap or abandons it.

4. Evaluation Protocol

We evaluate the workflow with a blind human preference study. The study compares multiple roadmap variants generated for the same research idea and the same input description. Holding the topic fixed across variants controls for topic preference and isolates roadmap-quality judgments from idea preference. The study design is illustrated in Figure 4.

4.1. Materials: Roadmap Variants

For each research idea we produce four roadmap variants, labeled A, B, C, and D. Variants A, B, and D are generated by three different unmodified ChatGPT versions, each prompted in an unstructured way (a single free-form instruction asking for a research roadmap on the given idea, with no schema and no review loop). Variant C is produced by our workflow: schema-constrained generation, optional reference templates, asynchronous compilation to a reviewable PDF artifact, and one or more human-review iterations through the revision_requested state described in Section 3.3.

The four variants are formatted identically before they reach an evaluator. Provenance metadata, generator name, model version, and any other identifying header is stripped, so an evaluator sees only the roadmap content under the labels Roadmap A, Roadmap B, Roadmap C, and Roadmap D. The mapping between labels and generators is held fixed within the study; we did not randomize letter order across evaluators in this round, so the labels are consistent but the labels themselves carry no provenance information.

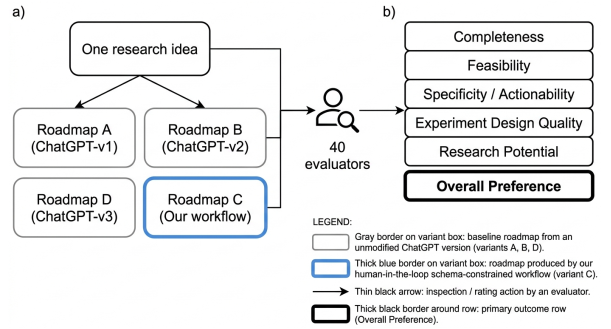

Figure 4. Study design. For one research idea, four anonymized roadmap variants are produced (A, B, D from three different unmodified ChatGPT versions; C from our human-in-the-loop schema-constrained workflow). Forty evaluators rate every variant on five quality dimensions and record a single overall preference. The overall preference row is the primary outcome.

4.2. Participants

We recruited 40 evaluators drawn from a research bootcamp cohort, composed of students, mentors, and adjacent practitioners. Every evaluator inspected all four roadmap variants for the shared research idea. Evaluators completed the survey independently and were not given the identity of any generator before submitting their ratings.

4.3. Rated Dimensions and Survey Instrument

Each evaluator rates every variant on five dimensions chosen for executability rather than fluency or helpfulness:

- Completeness. Does the roadmap cover research questions, methodology, datasets, baselines, metrics, milestones, risks, and deliverables?
- Feasibility. Can the roadmap be executed within the proposed timeline, given accessible datasets or tools, realistic milestones, and achievable deliverables?
- Specificity / Actionability. Does the roadmap make it clear what to read, build, test, compare, and submit?
- Experiment Design Quality. Does the experimental plan have suitable baselines, measurable metrics, ablations, a validation strategy, and clear success criteria?
- Research Potential. Does the roadmap have a clear problem, plausible novelty, meaningful evaluation, and a realistic minimum viable claim, such that the project could plausibly support a paper, workshop submission, or technical report?

For each dimension, the evaluator selects one variant (out of four) as the strongest. After completing the per-dimension items, the evaluator records a single overall preference: which roadmap they would actually adopt and execute. The overall preference is the primary outcome of the study. Evaluators also provide free-text feedback on why they selected the strongest roadmap, which roadmap they considered weakest, and what changes they would request before using the roadmap.

4.4. Analysis Plan

The primary statistic is the count of overall-preference votes received by each variant, out of 40. We test the null hypothesis of uniform preference across the four variants ($p_0 = 0.25$ per variant) with a one-sided binomial test on the count of votes received by the top-ranked variant. Per-dimension counts are reported when available; in the present round, per-dimension counts were collected at lower granularity than the overall preference vote, so we treat the dimension-level evidence as qualitative and rely on the free-text justifications to characterize why evaluators preferred or rejected a given variant. Section 5 reports the results, and Section 7 discusses the threats to validity that follow from this design.

5. Results

Study Setup and Variants

We collected human preference judgments from 40 evaluators comparing four anonymized roadmap variants for the same research idea, labeled A, B, C, and D. Variants A, B, and D were generated by three different unmodified ChatGPT versions using a single unstructured prompt per idea, while Variant C was produced by our human-in-the-loop, schema-constrained workflow. Each evaluator inspected all four roadmaps blind to provenance and rated them on the five dimensions defined in Section 4: completeness, feasibility, specificity and actionability, experiment design quality, and research potential. After scoring the per-dimension items the evaluator was asked which roadmap they would adopt overall.

Overall Preference

The headline result is concentrated and one-sided. Of the 40 evaluators, 34 (85%) selected Variant C, the roadmap produced by our workflow, as the one they would actually use to scope and execute the project. The remaining 6 (15%) preferences were distributed among the three unstructured-prompt baselines A, B, and D. Figure 5 visualizes the split, and Table 1 reports the count and share for each variant. A simple majority test rejects the null of uniform preference among the four variants ($p < 10^{-3}$ under a one-sided binomial test against $p_0 = 0.25$).

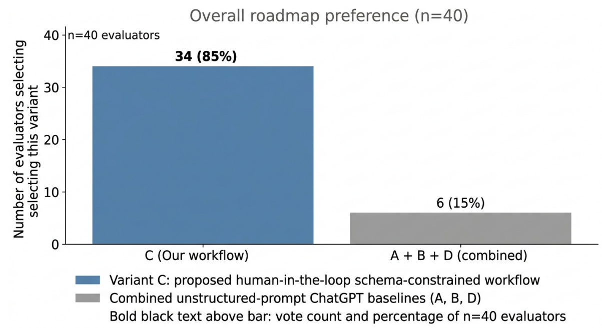

Figure 5. Overall roadmap preference across 40 evaluators. Variant C (the human-in-the-loop, schema-constrained workflow) is selected by 34 of 40 evaluators (85%); the three unstructured ChatGPT baselines A, B, and D collectively receive 6 votes (15%).

Per-Dimension Behavior

Across the five rated dimensions, completeness, feasibility, specificity and actionability, experiment design quality, and research potential, our impression from inspecting evaluator comments at the time of the survey is that the structured roadmap's explicit components (research questions, baselines, metrics, ablations, weekly milestones, and a stated minimum viable claim) were the most frequently cited reason to prefer Variant C, and that the unstructured baselines were most often criticized for missing or under-specified baselines and metrics, vague milestones, and overconfident contribution claims. We deliberately frame this as a researcher impression rather than a finding because the survey instrument did not retain free-text answers at the granularity required for direct quotation or reproducible qualitative coding; we note this as a limitation in Section 7. We did not record per-dimension vote counts at sufficient granularity to report a separate winner for each dimension; we therefore restrict the quantitative claim to the overall preference and treat the dimension-level evidence as qualitative.

What the Numbers Do and Do Not Show

The 34/40 preference is a measurement of evaluator-perceived roadmap quality at generation time, not of downstream project success. The result is consistent with our hypothesis that schema-constrained generation paired with human review produces roadmaps that are more complete and more actionable than one-shot prompting, but it does not establish that students

Variant	Generator	Overall preference (of 40)
A	ChatGPT, unstructured prompt	
B	ChatGPT, unstructured prompt	6 combined (15%)
D	ChatGPT, unstructured prompt	
C (ours)	Schema + human-review workflow	**34** (**85**%)

Table 1. Overall roadmap preference across 40 evaluators. Variant C is the output of the proposed human-in-the-loop schema-constrained workflow; A, B, and D are roadmaps produced by three different unmodified ChatGPT versions for the same idea. Lower-preference variants are reported as a combined count because individual A/B/D shares were not separately captured in the survey instrument.

who follow Variant C produce stronger final papers, finish milestones faster, or require fewer mentor interventions. Section 7 discusses these threats to validity in detail. We also note that all four variants were generated for a fixed set of research ideas drawn from a single bootcamp cohort, so the absolute share favoring Variant C should be read as an effect size on this evaluator population rather than a population-wide estimate.

6. Discussion

Research roadmap curation sits between brainstorming and execution. It is not the same as idea generation, literature review, code generation, or paper writing; it is the stage where an idea becomes operational. Many discussions of LLMs in research focus on whether models can write papers, generate hypotheses, or act as autonomous scientists. Our framing is narrower and more practical. We do not claim that an LLM can determine scientific truth or independently supervise research. We find that an LLM, when constrained by a schema and placed inside a human review loop, can help produce better first-pass research plans, and the 34/40 overall preference for the workflow's roadmap is consistent with that finding.

The model-agnostic design matters in practice. The contribution is not tied to a specific backend; different bootcamps, labs, or research programs can instantiate the same workflow using different LLMs, templates, review policies, or output formats. What stays fixed is the structure: schema-constrained generation, reviewable artifacts, an explicit revision-or-approve decision point, and a human accountable for the final plan. The asynchronous job lifecycle in Section 3.3 is one realization of that structure; others are possible, including synchronous web interfaces, terminal workflows, and editor extensions, as long as the revision_requested state remains a first-class concept.

7. Limitations

The study has several limitations that bound the strength of the 34/40 result.

First, human preference at generation time is not downstream project success. Evaluators who select Variant C are reporting which roadmap they would adopt, not which roadmap leads to the strongest final paper, the fastest milestone completion, or the smallest number of mentor interventions. The next round of work needs to track those downstream outcomes directly.

Second, the evaluation depends on evaluator expertise. Beginners may prefer detailed roadmaps that experts find unrealistic, while experts may penalize roadmaps for missing domain-specific subtleties. Our 40-evaluator cohort was drawn from a research bootcamp and includes students, mentors, and adjacent practitioners; the absolute share favoring Variant C should therefore be read as an effect size on this population rather than a population-wide estimate.

Third, the same roadmap schema may not work equally well across domains. A roadmap for reinforcement learning, scientific machine learning, multimodal retrieval, or AI-systems work may require different levels of detail and different default components. The schema in Figure 2 is intentionally coarse-grained, but a domain-specific instantiation may push more components into the methodology family.

Fourth, the workflow may inherit biases from the reference templates that condition generation. If the templates emphasize certain styles of project, the generated roadmaps may overproduce similar structures. We did not control for this in the present study.

Fifth, the labels A, B, C, and D were held fixed across evaluators rather than randomized per evaluator, so any latent label bias (for example, evaluators tending to read C more carefully because it sits in the middle of the list) would inflate the apparent preference for C. A future round should randomize label-to-variant mapping per evaluator and report the same primary outcome under that design.

Sixth, the survey instrument captured the overall preference at higher granularity than the per-dimension votes; this is why Section 5 reports the dimension-level evidence as qualitative rather than as five separate counts. A revised survey should record per-dimension votes at the same granularity as the overall preference so each dimension can be tested independently.

Seventh, the study uses a single shared research idea across all four roadmap variants. Holding the topic fixed controls for idea preference within the comparison, but it also means the 34 of 40 overall preference reported here is conditional on that one idea. The effect size could be inflated, attenuated, or reversed for ideas that are narrower, more domain-specific, or more poorly matched to the workflow's reference templates. A k-idea generalization, with the same blind protocol applied to multiple ideas drawn from different subareas, is required before the result can be read as a population claim about idea-to-roadmap conversion in general. We acknowledge this as the most consequential limitation of the present round.

8. Conclusion

We presented a human-in-the-loop LLM workflow for research roadmap curation. The workflow converts vague research ideas into structured, reviewable execution plans containing research questions, literature directions, datasets, baselines, metrics, ablations, risks, milestones, deliverables, and a minimum viable claim, and it places a human reviewer at an explicit revision_requested decision point rather than only at the prompt. In a blind preference study with 40 evaluators rating four anonymized roadmap variants for one shared research idea, 34 (85%) selected the workflow's roadmap as the one they would adopt, against 6 for the three unstructured ChatGPT baselines. The single-idea design means the absolute share should be read as conditional on the chosen idea, not as a population-wide estimate.

The result is consistent with, but does not by itself establish, the broader claim that schema-constrained generation under human review produces more usable first-pass research plans than one-shot prompting. Two natural follow-ups are most useful. First, downstream evaluation: track whether participants who execute a Variant-C roadmap actually finish more milestones on time, write stronger reports, or require fewer mentor interventions than those who follow an unstructured roadmap. Second, domain-specific schemas: adapt the twelve-component schema to the conventions of specific research subareas and re-run the preference study against each domain's expert evaluators. The value of LLMs in early-stage research planning is not autonomous research generation, but structured assistance under human oversight, and we expect the operational benefits to grow as the structure and the oversight become more specific to the domain.

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